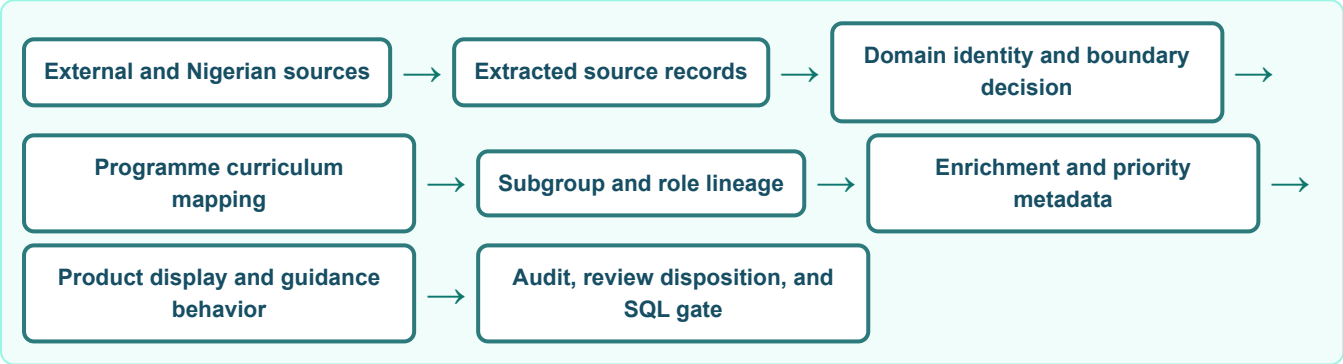
Sifter Career Taxonomy — Academic Methodology

Document ID: `METHODOLOGY-2026-09-01`
Status: working methodology and reproducibility handoff
Scope: domain identity, Nigerian programme-to-domain mapping, subgroups, roles, enrichment, and evidence boundaries

0. How to read this document

This document is organised as a traceable chain. Every important decision has a stable key that can be searched in the audit JSON files and followed into the product data path.



▶ View diagram definition

Stage	Stable key	Authoritative detail
Source	`SRC-*`	§1 Source register
Domain identity	`DOM-*`	§2 Domain derivation
Programme mapping	`MAP-*`	§3 Programme-to-domain evidence
Entity enrichment	`ENT-*`	§4 Subgroup, role, and Nigerian context
Review disposition	`REV-*`	§5 Review closure
Verification	`CHK-*`	§6 Reproducibility and release gate

1. Source register

Source key	Canonical source/version	What it supports	What it does not prove	Repository/audit detail
`SRC-ISIC`	UNSD ISIC Rev.5, with Rev.4 correspondence retained	Industry context, activity boundaries, completeness survey	Occupations, skills, programme suitability, or Nigerian demand	ISIC crosswalk

Source key	Canonical source/version	What it supports	What it does not prove	Repository/audit detail
`SRC-ISCO`	ILO ISCO-08	Occupational hierarchy, work-function and skill-level anchors	Sifter domain names, local demand, or learner readiness	ISCO crosswalk
`SRC-ESCO`	ESCO v1.2.1 classification archive	Normalised occupation labels, aliases, URIs, descriptions, and skills	Nigerian licensing or demand	Source traceability
`SRC-ONET`	O*NET 31.0	Tasks, tools, work activities, preparation, job zones, and skill proxies	Nigerian requirements or demand	O*NET reconciliation
`SRC-ILO`	ILO/ILOSTAT	International occupational and labour-statistics context	A Nigerian role or vacancy without a Nigeria-specific series	Source-layer review
`SRC-WEF`	WEF Future of Jobs 2025	Global trend, emerging-role, and future-skill signals	A complete occupation list or Nigerian market demand	WEF normalization
`SRC-CCMAS`	NUC/CCMAS extracts and JAMB-linked programme data	What a Nigerian programme teaches and its stated outcomes	Automatic eligibility beyond the applicable JAMB/university rules	CCMAS reviewed links
`SRC-NG`	Nigerian regulators, professional bodies, universities, and checked local sources	Nigerian terminology, institutions, regulation, credentials, curriculum, and directly shown local signals	A local demand claim when the source does not show demand	Nigerian coverage

Source registration is not ingestion proof. A published claim must point to the source record, file, page, section, URL, or exact extracted row that supports it. The complete 96-entry source catalog, all JAMB/CCMAS source-file names, and available source hashes, and layer-by-layer source ownership are reproduced in the standalone source and behavior inventory. The academic handoff packet must distribute that inventory with this document; the links below are the public source locations and the inventory identifies which local source files were actually checked.

Complete source-family register for the academic packet

The following register is reproduced here so this methodology remains useful when read without repository access. It is the source-family register used for domain, subgroup, role, skill, competency, tool, certification, resource, and Nigerian-enrichment routing. A routed source is not automatically an extracted fact; the evidence state is recorded separately.

ID	Source family	Public reference
`global_esco`	ESCO occupations and skills	ESCO
`global_onet`	O*NET 31.0 Database	O*NET
`global_isco`	ILO ISCO-08	ISCO
`global_wef`	World Economic Forum Future of Jobs 2025	WEF

ID	Source family	Public reference
`global_sfia`	SFIA 9	SFIA
`global_nice`	NIST NICE Workforce Framework	NICE
`global_supply_chain`	ASCM supply-chain body of knowledge	ASCM
`global_cips`	CIPS Global Standard for Procurement and Supply	CIPS
`global_project_management`	Project Management Institute standards	PMI
`global_health`	WHO health workforce resources	WHO
`global_fao`	FAO agriculture and food-systems resources	FAO
`global_woah`	WOAH animal-health standards	WOAH
`global_icao_iata`	ICAO and IATA aviation standards	ICAO / IATA
`global_imo_imarest`	IMO and IMarEST maritime standards	IMO / IMarEST
`global_built_environment`	RICS built-environment and property standards	RICS
`global_energy`	SPE petroleum-engineering competency resources	SPE
`global_nuclear`	IAEA nuclear and radiation resources	IAEA
`global_unesco`	UNESCO education, science, and culture resources	UNESCO
`global_creative`	ScreenSkills creative-industry role profiles	ScreenSkills
`global_game_dev`	International Game Developers Association resources	IGDA
`tesda_psf_game_development`	Philippine Skills Framework for Game Development	TESDA
`global_xr`	Khronos Group open standards	Khronos
`qt_cfqt`	European Competence Framework for Quantum Technologies v3.0	Quantum Flagship
`nac_arts_technical_theatre`	Singapore National Arts Council technical-theatre framework	Skills Framework
`skills_england_live_event`	Skills England Live Event Technician standard	Skills England
`mqa_performing_arts`	Malaysian Qualifications Agency Performing Arts standards	MQA
`jff_xr`	JFF XR and Immersive Learning career pathways	JFF
`arm_robotics_career`	ARM RoboticsCareer competency pathways	RoboticsCareer
`skills_england_robotics`	Skills England Robotics Engineer maps	Robotics

ID	Source family	Public reference
`british_esports`	British Esports career pathways	British Esports
`pearson_esports`	Pearson BTEC Esports career pathways	Pearson
`global_esports`	Hitmarker esports and gaming jobs	Hitmarker
`global_music`	IFPI global music-industry resources	IFPI
`global_wipo`	WIPO intellectual-property resources	WIPO
`global_cfa_finance`	CFA Institute investment-profession resources	CFA
`global_garp_risk`	Global Association of Risk Professionals	GARP
`global_driver_training`	DVSA National Standard for Driver and Rider Training	DVSA
`ng_jobberman`	Jobberman Nigeria jobs	Jobberman
`ng_myjobmag`	MyJobMag Nigeria feeds	MyJobMag
`ng_nbs`	National Bureau of Statistics Labour Force Survey	NBS
`ng_national_ai`	Nigeria National Artificial Intelligence Strategy 2025	Local checked source document
`ng_nbte`	NBTE National Occupational Standards and curricula	NBTE
`ng_cpn_syllabus_2021`	CPN 2021 Examination Syllabus	CPN syllabus
`ng_ncs`	Nigeria Computer Society	NCS
`ng_nsqf_nos`	NBTE NSQF and National Occupational Standards	NSQF
`ng_cieeen`	Council for Innovative Engineering and Emerging Engineering	CIEEEN
`ng_health_ministry`	Federal Ministry of Health and Social Welfare	FMoHSW
`ng_mls`	Medical Laboratory Science Council of Nigeria	MLSCN
`ng_clmis_corbon`	CORBON construction labour-market information and NOS	CORBON
`ng_qsrbn`	Quantity Surveyors Registration Board of Nigeria	QSRBN
`ng_nnra`	Nigerian Nuclear Regulatory Authority	NNRA
`ng_trcn`	Teachers Registration Council of Nigeria	TRCN
`ng_nogicjqs`	Nigerian Oil and Gas Industry Content Joint Qualification System	NOGIC JQS
`ng_cmd`	Centre for Management Development Nigeria	CMD

ID	Source family	Public reference
`ng_nasrda`	National Space Research and Development Agency	NASRDA
`ng_nema`	National Emergency Management Agency	NEMA
`ng_fmacce`	Federal Ministry of Arts, Culture and Creative Economy	FMaCCE
`ng_ican_syllabus`	Institute of Chartered Accountants of Nigeria 2026 syllabus	ICAN
`ng_cibn`	Chartered Institute of Bankers of Nigeria competency framework	CIBN
`ng_nuj`	Nigeria Union of Journalists professional ethics	NUJ
`ng_nla`	Nigerian Library Association and LRCN context	NLA
`ng_nicfost`	Nigerian Council of Food Science and Technology	NICFOST
`ng_nihotour`	NIHOTOUR RIVAC registration portal	RIVAC
`ng_fmafs`	Federal Ministry of Agriculture and Food Security	FMAFS
`ng_npa_psychology`	Nigerian Psychological Association	NPA
`ng_fashion_council`	Nigerian Fashion Council and FADAN	FADAN
`ng_sports_context`	Nigeria Olympic Committee sports-sector context	NOC
`ng_public_health_context`	NPHCDA and Nigerian public-health sources	NPHCDA
`ng_nemet`	Nigeria Meteorological Agency (legacy registry key)	NiMet
`ng_nimet`	Nigerian Meteorological Agency	NiMet
`ng_nmdpra`	Nigerian upstream and midstream petroleum regulators	NUPRC / NMDPRA
`ng_legal_context`	Nigerian legal, immigration, diplomacy, and IP source set	NBA
`ng_cipsmn`	Chartered Institute of Purchasing and Supply Management of Nigeria	CIPSMN
`ng_nas_research`	Nigerian Academy of Science and national research sources	NAS
`ng_nuc`	National Universities Commission CCMAS and standards	NUC
`ng_cpn`	Computer Professionals Registration Council of Nigeria	CPN

ID	Source family	Public reference
`ng_telecom`	NCC and NITDA ICT ecosystem sources	NCC / NITDA
`ng_engineering`	COREN and Nigerian Society of Engineers	COREN / NSE
`ng_built_environment`	Nigerian built-environment regulators and bodies	ARCON / SURCON
`ng_health`	Nigerian health-profession regulators	MDCN / NMCN
`ng_finance`	Nigerian finance and accounting bodies/regulators	SEC / CBN
`ng_business`	CIPM Nigeria and Nigerian Institute of Management	CIPM
`ng_supply_chain`	CILT Nigeria and Nigerian supply-chain bodies	CILT Nigeria
`ng_legal`	Nigerian Bar Association and legal regulators	NBA
`ng_education`	NUC, NCCE and Nigerian education standards	NCCE
`ng_agriculture`	Nigerian agriculture, food and veterinary authorities	NAFDAC / VCN
`ng_energy`	Nigerian upstream, midstream and downstream energy regulators	NUPRC
`ng_aviation_maritime`	Nigerian aviation and maritime authorities	NCAA / NIMASA
`ng_creative`	Nigerian creative-industry and media sources	NFVCB / NUJ
`ng_gaming`	National Lottery Regulatory Commission and gaming ecosystem	NLRC
`ng_environment`	NESREA and Nigerian environmental regulators	NESREA
`ng_surcon`	Surveyors Council of Nigeria	SURCON
`ng_mdcn`	Medical and Dental Council of Nigeria	MDCN
`ng_transport`	Federal Ministry of Transportation	Transportation Ministry
`ng_aviation`	Federal Ministry of Aviation and Aerospace Development	Aviation Ministry
`ng_choice_game_developer_2026`	Choice Talents Nigeria game-developer vacancy	Dated listing

The source registry also contains the local JAMB brochure set and the original CCMAS PDF set. The exact filenames are listed in the source and behavior inventory, alongside hashes and the distinction between a checked source document, a checked extract, and a registry-only route.

For a standalone academic copy, the primary Nigerian document inputs are also listed here.

JAMB/IBASS brochure files: `Administration.pdf`, `Agriculture.pdf`, `Arts _ Humanities.pdf`, `Education.pdf`, `Engineering _ Enviromental _ Technology.pdf`, `Law.pdf`, `Medica_Pharmaceutical Health Science.pdf`, `Sciences.pdf`, `Social Management Science.pdf`, and `Specific Entry Requirements.pdf`.

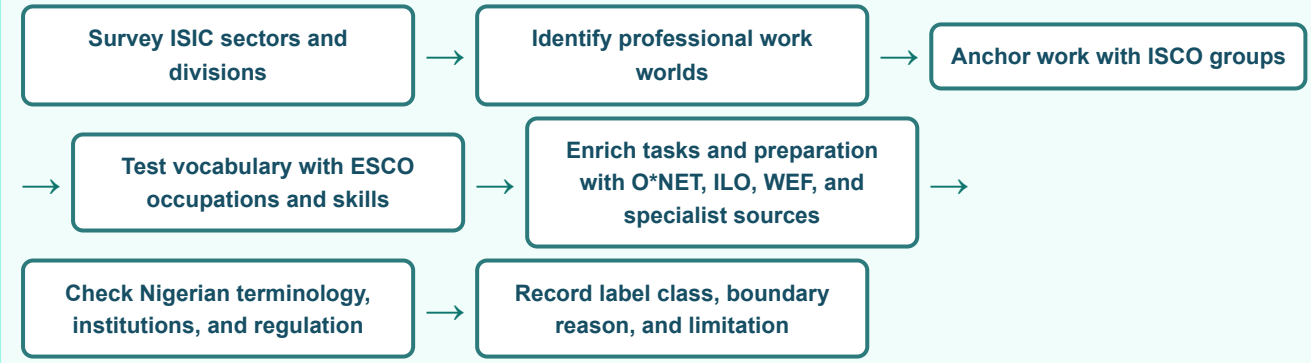
Original CCMAS/NUC files:

```
Administration-and-Management.pdf
Agriculture-2023.pdf
Allied-Health-Sciences-2023.pdf
Architecture-CCMAS-2023-FINAL.pdf
Arts-CCMAS-2023-FINAL.pdf
Basic-Medical-Sciences-CCMAS-FINAL-December-26-2022.pdf
Communication-and-Media-Studies-CCMAS-2023-FINAL.pdf
Computing-CCMAS-2023-FINAL.pdf
Education-CCMAS-2023-New.pdf
Engineering-Technology-CCMAS-2023-FINAL.pdf
Environmental-Sciences-CCMAS-2023-FINAL.pdf
Law-ALL.pdf
Medicine-and-Dentistry-CCMAS-2023-FINAL.pdf
Pharmacy-and-Pharmaceutical-Sciences-CCMAS-2023-FINAL.pdf
Sciences-CCMAS-2023-FINAL.pdf
Social-Sciences-CCMAS-FINAL-2023-A.pdf
Veterinary-Medicine-CCMAS-2023-FINAL.pdf
```

The original documents are the authoritative material; the checked-in JSON and text files are extraction and audit artifacts that preserve the relevant course codes, source text, outcomes, and locators.

2. Domain derivation and identity

2.1 Derivation flow



▶ View diagram definition

The domain library is a product ontology, not a replacement for an official classification. ISIC provides the economic completeness anchor; ISCO, ESCO, O*NET, ILO, WEF, specialist frameworks, and Nigerian evidence support different occupational or contextual claims.

2.2 Domain identity row contract

Row key	Required fields	Decision question	Downstream links
`DOM-D001` … `DOM-D166`	Sifter ID, current label, source-library label, cluster, ISIC raw values, ISCO raw values, ESCO role count, label class, boundary reason, limitation	Is this exact source terminology, a source-supported umbrella, or a curated addition?	`domain_meta.json`, `domain_structure.json`, programme links, subgroup/role rows

Row key	Required fields	Decision question	Downstream links
`DOM-D167`, `DOM-D168`	Same fields plus later-addition source registry and limitation	Is the later addition in scope and supported without pretending it is an official ISIC/ISCO label?	Same downstream path, with curated status visible

2.3 Boundary classes

Class key	Meaning	Minimum defensible basis	Product language
`DOM-EXACT`	The label occurs as the source's own terminology	Exact source label plus source locator	“This name is used by the source.”
`DOM-UMBRELLA`	A practitioner-facing label groups source-recognised occupations	Shared learning foundation, entry route, outputs, tools, employer context, and coherent occupational anchors	“This is a Sifter umbrella for related source occupations.”
`DOM-CURATED`	A product addition closes a documented coverage gap	Supporting source records, explicit boundary rationale, and limitation	“This is a curated product category, not an official classification.”

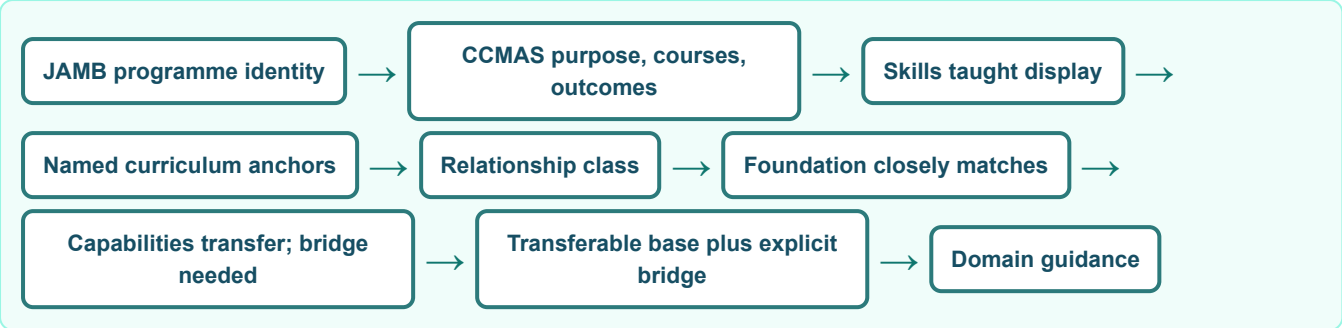
2.4 Current domain identity result

Check key	Result	Evidence
`DOM-COUNT`	168 domain records; 160 visible in the current product scope	`domain_meta.json` and domain traceability JSON
`DOM-EXACT`	7	Naming provenance audit
`DOM-UMBRELLA`	159	Same audit
`DOM-CURATED`	2 later additions, D167–D168	Same audit
`DOM-ISCO-RAW`	8 raw ISCO values absent from the supplied workbook	ISCO crosswalk; parent/sibling candidates are recorded without silent replacement

The eight raw ISCO values are a provenance/code-quality boundary, not evidence that the associated domains are invalid. The raw value and its disposition must remain visible until the original source meaning supports a canonical correction.

3. Programme-to-domain evidence

3.1 Mapping flow



▶ View diagram definition

3.2 Mapping row contract

Row key	Nigerian evidence required	Decision	Student-facing explanation	Stored detail
`MAP-DIRECT`	At least two specific CCMAS course/objective anchors closely support the domain foundation	Direct	“Your programme teaches a foundation closely related to this domain.”	Course code, title, outcome/content text, source locator, domain anchor, review state
`MAP-FUNCTIONAL`	At least two specific CCMAS course/objective anchors support capabilities used in the domain	Functional	“Your programme gives you useful preparation, but you will need additional domain-specific capability.”	Same evidence fields plus bridge explanation
`MAP-ADJACENT`	At least one domain-specific prerequisite plus relevant transferable P7/P8 evidence	Adjacent	“This is a possible pivot. Your current foundation helps, but you need this bridge before specialising.”	Same evidence fields plus explicit bridge requirement

“Strong” means named, source-backed curriculum anchors support the relationship. It is not a similarity score, employment prediction, or guarantee of competence. Global occupational evidence can explain the target domain, but cannot repair a weak Nigerian programme-to-domain link.

3.3 The 153 and remaining CCMAS occurrences

The 711 candidate-link occurrences were reviewed as follows:

Review key	Count	Final disposition	Product/seed rule
`REV-CCMAS-CERTIFIED`	547	Certified Nigerian curriculum link	May publish as the applicable relationship class with source evidence
`REV-CCMAS-BRIDGE`	163	Explicit Nigerian bridge link	Publish as guided preparation; never present as automatic qualification
`REV-CCMAS-REMOVED`	1	Unsupported candidate removed	Do not seed or display as an active mapping
`REV-CCMAS-FOLLOWUP`	0	No current follow-up records	Preserve the review artifact and rerun the audit after any source change

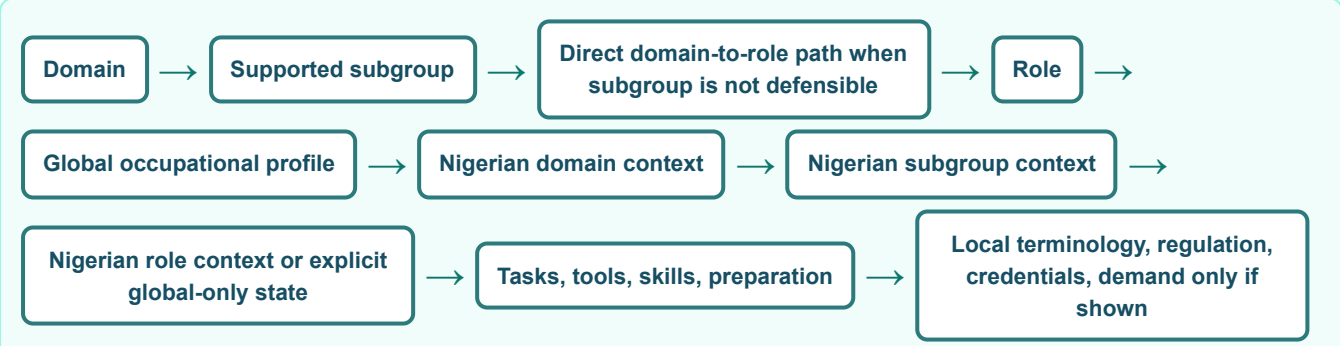
The earlier **153 insufficient** count was a before-state diagnostic. It is not an unresolved queue: those occurrences are represented in the final certified/bridge/removed dispositions above.

3.4 Downstream impact contract

Impact key	What is checked	Current result	Authoritative artifact
`IMPACT-ID`	Domain IDs and ownership remain stable	No invalid domain references	Downstream impact audit
`IMPACT-LABEL`	Display labels match canonical IDs	No label mismatches	Same audit
`IMPACT-PROGRAMME`	Programme-domain links remain connected	1,683 programme links	Same audit
`IMPACT-JAMB`	Reviewed JAMB overlay remains connected	3,163 active JAMB links; 710 reviewed links	Same audit and reviewed JAMB JSON
`IMPACT-INHERITANCE`	Skills, tools, competencies, and certification inheritance remains attached by ID	No invalid downstream references reported	Same audit

4. Subgroup, role, and Nigerian context

4.1 Entity lineage flow



▶ View diagram definition

4.2 Entity row contract

Entity key	Global sources	Nigerian layer	Safe display behavior
`ENT-DOMAIN`	ISIC/ISCO/ESCO/O*NET/ILO/WEF as applicable	Local terminology, institutions, regulation, credentials, and local signals	Show domain meaning and portable foundation; separate global context from Nigerian evidence
`ENT-SUBGROUP`	ISCO/ESCO/O*NET or specialist framework	Local work-family evidence where supported	Explain what the specialization adds beyond the domain foundation; do not invent a subgroup for symmetry

Entity key	Global sources	Nigerian layer	Safe display behavior
`ENT-ROLE`	ESCO/ISCO/O*NET/WEF/specialist framework	Exact local role-family evidence, or explicit reviewed global-only state	Show work, outputs, preparation, progression, essential/optional items, sources, and limitation

If no defensible subgroup exists, use `domain → role`. This is an intentional product route and not a data failure.

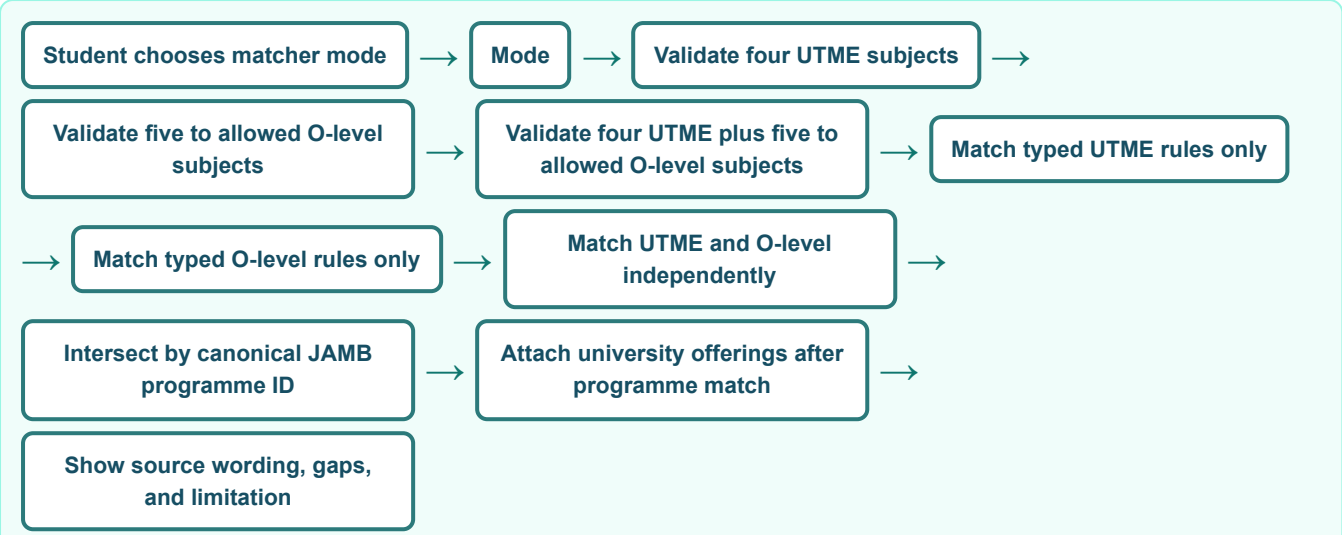
4.3 Current coverage snapshot

Check key	Full/current scope	Visible Nigerian scope	Interpretation
`ENT-SUBGROUP-PROVENANCE`	555/555	542/542	All visible subgroups have actual local enrichment
`ENT-ROLE-MERGED`	5,209	—	Every merged role has some source evidence
`ENT-ROLE-ANCHOR`	5,130 explicit source anchors; 5,090 work-profile anchors	—	Anchor types are retained separately
`ENT-ROLE-CONTEXT`	—	5,093/5,093 valid context	5,089 have actual local enrichment
`ENT-ROLE-GLOBAL-ONLY`	—	4	Snowboard instructor, ski instructor, snow-clearing worker, and hop farmer remain explicitly global-only; no direct Nigerian counterpart is asserted

“Actual local enrichment” means at least one source-backed local skill, tool, competency, credential, regulation, or role-family record. It does not mean the exact canonical title is a regulated Nigerian title or that demand is proven.

5. Admission subject-matching method

The admission matcher is a three-mode method, not one flattened subject list. The same canonical JAMB programme ID is the join key throughout.



▶ View diagram definition

- **JAMB-only:** Use exactly four normalized UTME subjects, including Use of English. Match required items and one-of groups in typed `utme` records only.
- **O-level-only:** Use five to the configured early-guidance maximum O-level subjects. Match typed `olevel` records only, including named credits and the total minimum where the source supplies it.
- **Combined:** Match the two layers independently, intersect only on the same canonical JAMB programme ID, and preserve O-level missing subjects or an unparsed O-level state. Do not count a UTME match as full eligibility.

The API and UI fail closed when the subject catalogue, structured options, and typed rule records disagree. Unparsed brochure prose remains inspectable on programme detail but does not become an exact subject match. After a canonical programme is selected, university offerings supply address, requirements, and notes; they do not create a new programme version. The behavior is implemented in `backend/routes/student_pipeline_routes.py` and `src/screens/careerIntelligence/AdmissionBrowseScreen.tsx`, and independently checked by `backend/tests/test_student_pipeline_match.py` and `docs/manual-review/subject-selection-matrix-run-2026-08-18.md`.

6. Worked source-to-rule examples

These examples show the level of explanation required for every published relationship. Each example names the source fact, the rule applied, the link to the repository evidence, and the reason the result is safe to display.

6.1 Domain identity: D001 Software Engineering

Stage	Evidence or decision
Source fact	The supplied domain-definition library has `D001 · Software Engineering` in its Information & Communication Technology cluster, with a defined boundary based on shared work outputs, learning foundation, entry routes, and occupational anchors.
Industry rule	ISIC is used as an industry-context completeness check, not as the literal domain name. The trace record preserves the library's raw `K620` and `K630` values and records that they require revision/granularity crosswalk review in the local Rev.5 workbook.

Stage	Evidence or decision
Occupational rule	The library anchors the domain to ISCO sub-major 25 and unit groups `2512`, `2513`, `2514`, and `2519`; the local ISCO workbook validates those unit groups. ESCO is then filtered under those anchors for normalized occupation and skill evidence.
Repository link	domain traceability JSON, domain `D001`; ISIC crosswalk; domain metadata.
Rationale	"Software Engineering" is not claimed to be an official ISIC label. It is an exact label in the supplied Sifter library, supported by a coherent occupational anchor and linked role vocabulary. The invalid/missing ISIC values stay visible until the canonical Rev.5 crosswalk is frozen; no code is silently replaced.

This example is defensible because it distinguishes label provenance from code validation. It does not pretend that an ISIC code automatically generated the human-facing label.

6.2 Domain identity and occupational boundary: D023 Medicine

Stage	Evidence or decision
Source fact	The library record is `Medicine & Clinical Practice`, displayed in the product as the source-supported umbrella `Medicine`, with Health & Biomedical as its cluster.
Industry rule	The record retains the raw health-activity anchors `R861`, `R862`, and `R863` for an explicit ISIC crosswalk; the audit records that `R863` is not found as a direct value in the supplied local Rev.5 workbook.
Occupational rule	The record includes ISCO sub-major 22 and medical unit-group anchors. The official ISCO-08 structure identifies `2211 Generalist Medical Practitioners` and `2212 Specialist Medical Practitioners`; the supplied local workbook does not contain raw `2213`-`2216`, so those raw values remain review findings rather than invented specialist codes.
Nigerian curriculum rule	The reviewed JAMB/CCMAS link for `Medicine and Surgery (Human Medicine)` points to D023 as direct and retains named course anchors such as Introductory Physiology, History of Medicine, Multidisciplinary Healthcare, and Introductory Pharmacology.
Repository link	D023 traceability record; reviewed JAMB/CCMAS links; downstream impact audit.
Rationale	"Medicine" is a usable display umbrella because the source library explicitly groups clinical practice and the occupational anchors support the work family. The product must still distinguish doctors, technical health roles, public health, and biomedical pivots at subgroup/role level; it must not claim that every role is a doctor or that a programme guarantees licensure.

The official ISCO-08 PDF gives an independent locator for the occupational fact: Volume 1, printed PDF page 36 lists the Health Professionals structure, and the unit-group definition for `2211` appears on printed pages 135-136. The repository crosswalk is the controlling record for the absent raw values.

6.3 Programme-to-domain mapping: Computer Science to Software Engineering

Stage	Source fact -> rule -> link -> rationale
Source fact	`Computing-CCMAS-2023-FINAL.pdf`, extracted as `data/career-mapping/_ccmas_txt/computing.txt`, B.Sc. Computer Science overview, printed page 27, says the programme covers computer programming, networking, algorithms, computing systems, and software development. Its objectives include a broad foundation, problem solving with appropriate tools, software/algorithm design, teamwork, communication, and lifelong learning.
Rule	A Direct relationship requires at least two specific Nigerian CCMAS course or objective anchors closely supporting the destination domain foundation.
Applied link	The reviewed map links `Computing::Computer Science` to `D001 Software Engineering` using `COS 201 Computer Programming I`, `COS 202 Computer Programming II`, and `SEN 201 Introduction to Software Engineering`; the occurrence is `certified_nigerian_curriculum_link`.
Rationale	The mapping is not based on the words “Computer Science” alone. The named programming and software-engineering courses are direct curriculum evidence. The domain card can therefore explain the connection, while still saying the student may need role-specific practice and evidence.

The same programme maps functionally to Data Science using mathematics, statistics, and project-management anchors, and to Robotics Engineering using mathematics, physics, software engineering, and project-management anchors. Those are separate links because the curriculum supports useful capabilities without being the complete foundation for those occupations. See the active reviewed map under `Computer Science` and the CCMAS extracted source, printed pages 27-28 and 50.

6.4 Programme alias and Nigerian source handling: Criminology and Security Studies

Stage	Source fact -> rule -> link -> rationale
Source fact	The JAMB brochure audit finds the canonical name in the Arts/Humanities brochure page 126, Law brochure page 13, and Social Management Sciences brochure pages 1, 36, 39, 41, 42, 44, 45, 49, and 50. One catalogue row is printed under a Law faculty block.
Rule	Preserve the canonical JAMB programme identity; treat the faculty-placement wording as a declared display alias, not as a new programme. Preserve source variants when their requirements differ.
Applied link	The reviewed map links the programme directly to `D151 Criminology, Criminal Justice` using named CCMAS courses including Introduction to Criminology and Security Studies, Introduction to Criminal Justice System, Nigerian Law Enforcement and Security System, Criminal Procedure and Evidence, Forensic Science, and Intelligence and Investigation Management.
Rationale	This prevents a catalogue layout error from changing the academic identity. The student sees one programme, the source variants remain inspectable, and the destination domain is justified by actual course anchors rather than the word “Law” in a faculty block.

Evidence: `Social Management Science.pdf`, `Arts _ Humanities.pdf`, and `Law.pdf` in the local JAMB brochure set; JAMB brochure source audit, reviewed link record, and the active map.

6.5 The historical 153 and remaining bounded records

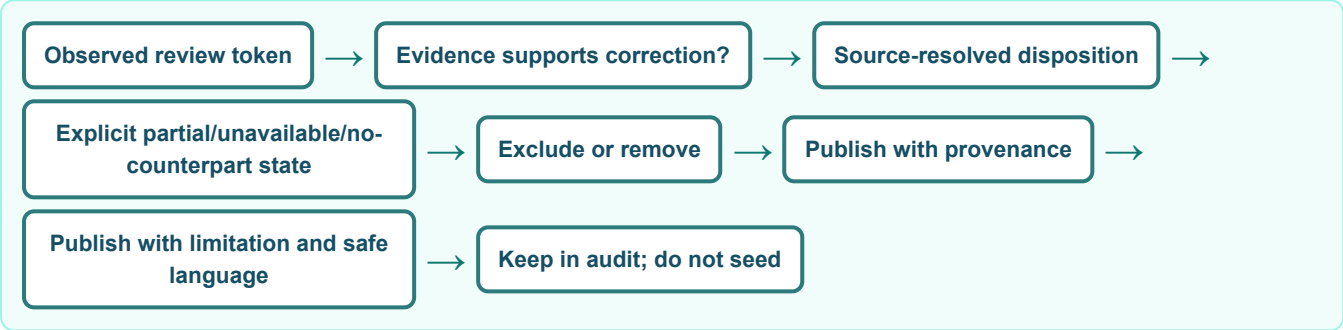
The earlier complement audit reported 547 sufficient, 153 insufficient, and 11 missing-anchor occurrences. Those numbers were diagnostic findings, not a final publication state. The reviewed artifact now accounts for all

711 occurrences: 547 certified, 163 explicit Nigerian bridge links, and one unsupported candidate removed. The bridge links are explanatory routes, not automatic qualifications. The eleven missing-anchor cases were not fabricated into exact links; the review either resolved them through a supported bridge or preserved the limitation.

The same rule applies to the remaining entity records. “No source candidate,” “partial definition,” “global-only,” “unavailable Nigerian evidence,” and “ambiguous trend signal” are valid bounded dispositions only when the audit stores the source search, decision, limitation, and safe display language. A reviewed record is not necessarily a complete record.

6. Review closure

6.6 Review-state flow



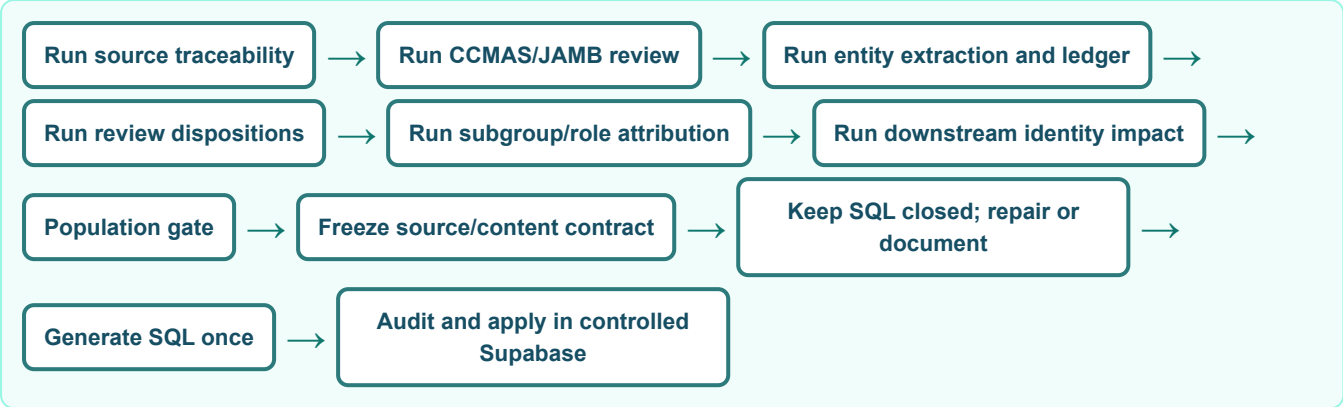
▶ View diagram definition

Review key	Count	Meaning
`REV-ALL`	1,538 dispositioned records	Every inventoried finding has an explicit treatment
`REV-FOLLOWUP`	0	No record remains marked for follow-up by its source ledger
`REV-PARTIAL`	220	Intentional partial source definitions, not fabricated completeness
`REV-VARIANTS`	445	Conflicting university/source variants preserved for institution-specific display
`REV-NO-CANDIDATE`	135	No source candidate explicitly preserved
`REV-WEF-UNMATCHED`	26	Retained as WEF signals; no role creation
`REV-WEF-AMBIGUOUS`	2	Retained as WEF signals; no role assignment
`REV-GLOBAL-ONLY`	4	Global role retained; no direct Nigerian counterpart asserted

Full detail: review-state dispositions and review-state inventory.

7. Reproducibility and release gate

7.1 Verification flow



▶ View diagram definition

7.2 Commands

```
python data/career-mapping/_audit_domain_source_traceability.py
python data/career-mapping/_build_programme_domain_evidence.py
python data/career-mapping/_audit_ccmas_jamb_inheritance.py
python data/career-mapping/_audit_ccmas_external_complements.py
python data/career-mapping/_build_nigerian_entity_extraction_audit.py
python data/career-mapping/_build_nigerian_entity_review_ledger.py
python data/career-mapping/_audit_review_state_inventory.py
python data/career-mapping/_audit_review_state_dispositions.py
python data/career-mapping/_audit_subgroup_role_source_traceability.py
python data/career-mapping/_audit_role_attribution_reconciliation.py
python data/career-mapping/_audit_domain_identity_downstream_impact.py
python data/career-mapping/_validate_nigerian_entity_population_gate.py
```

7.3 Current release status

The integrity checks pass: 5,795 audit entities reconcile to 5,795 ledger entities, with zero follow-up rows, zero provenance errors, zero invalid states, and zero unresolved context/no-direct rows. The exhaustive-search declaration has been reconciled to the current evidence, so the SQL gate is open for regeneration. This remains an honest release control: passing the population gate does not prove that every occupation has an exact Nigerian counterpart or demand signal. The newly generated SQL must still be transported, audited, and applied in a controlled deployment.

8. Claims that must not be made

- The taxonomy is an official ISIC, ISCO, ESCO, O*NET, or Nigerian classification.
- Global evidence proves Nigerian demand.
- A programme guarantees a career, competence, admission, or licensure.
- Every role is an exact Nigerian occupation or regulated title.
- A source URL alone is the same as extracted source content or a mapping decision.

The defensible claim is narrower: Sifter is a traceable product classification. Its relationships are supported by identified source records, documented rules, explicit limitations, and reproducible audit outputs.